

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

MINI PROJECT

RETAIL STORE SALES ANALYSIS AND VISUALIZATION USING PYTHON, POWER BI, AND TABLEAU

Framework for Data and Visual Analytics

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ABSTRACT

This mini project, carried out for the subject Framework for Data and Visual Analytics (FDVA), analyses a Retail Store Sales dataset using Python, Microsoft Power BI and Tableau. The dataset contains 200 transaction records with details such as sale date, city, product category, units sold, unit price, discount, payment method, sales channel and customer rating. Python with Pandas was used to handle missing values, standardise dates and numeric fields, and create derived measures such as net sales and discount percentage. Matplotlib and Seaborn were used to create eight charts, including category-wise transactions, city-wise sales, monthly sales trends and a correlation heatmap. Dashboard layouts for Power BI and Tableau demonstrate KPI cards, filters and charts.

In this sample dataset, total net sales are approximately ₹176,087 with groceries contributing the most transactions. The project demonstrates how preprocessing the data, visualisation and dashboards can can convert transaction data into useful retail business insights.

1. Data Preprocessing

The dataset (Retail_Sales_Data.csv) has 200 rows and 22 columns. It was cleaned in Python using Pandas, as shown below.

1.1 Loading and inspecting the data

```
df = pd.read_csv("Retail_Sales_Data.csv")
df.shape        # (200, 22)
df.info()
```

Sale dates, prices, discounts, units sold and customer ratings were inspected and converted to suitable data types where needed.

1.2 Missing values and duplicates

```
print(df.isnull().sum())
print("Duplicate rows:", df.duplicated().sum())
```

Customer_Rating (12), Payment_Method (5) and Stock_Status (4) contain missing values in this sample dataset. Duplicate rows should also be checked.

1.3 Fixing data types and filling missing values

```
df['Customer_Rating'] = pd.to_numeric(df['Customer_Rating'], errors='coerce')
df['Customer_Rating'] = df['Customer_Rating'].fillna(
    df['Customer_Rating'].median())
# Fill categorical missing values with the mode
df['Payment_Method'] = df['Payment_Method'].fillna(
    df['Payment_Method'].mode()[0])
```

Numeric fields were converted to numeric types; missing ratings were filled with the median, while missing categorical values were filled with the mode. This keeps the dataset usable for charting.

1.4 Standardising text values

```
df['City'] = df['City'].astype(str).str.strip().str.title()
df['Product_Category'] =
df['Product_Category'].astype(str).str.strip().str.title()
df['Sales_Channel'] = df['Sales_Channel'].astype(str).str.strip().str.title()
df['Payment_Method'] =
df['Payment_Method'].astype(str).str.strip().str.upper()
df['Promotion_Applied'] =
df['Promotion_Applied'].astype(str).str.strip().str.title()
```

Text labels were trimmed and standardised so that the same category or payment method is represented consistently in the analysis.

1.5 Converting the sale date

```
clean['Sale_Date'] = pd.to_datetime(
    clean['Sale_Date'].astype(str).str.strip(),
    errors='coerce')
clean['Year'] = clean['Sale_Date'].dt.year
clean['Month_Name'] = clean['Sale_Date'].dt.month_name()
```

The sale date was converted to a datetime field, enabling year and month extraction for time-based analysis. Invalid dates, if any, should be reviewed before dashboarding.

1.6 New columns and export

```
clean['Net_Sales_INR'] = clean['Gross_Sales_INR'] - clean['Discount_INR']
clean['Discount_Percentage'] = clean['Discount_INR'] /
clean['Gross_Sales_INR'] * 100
clean.to_csv("Cleaned_Retail_Sales.csv", index=False)
```

Derived fields such as Year, Month_Name, Net_Sales_INR and Discount_Percentage support the charts and dashboards. The cleaned dataset was exported for reuse in Python, Power BI and Tableau.

2. Data Visualization Using Python

Eight charts were created from the cleaned data using Pandas, Matplotlib and Seaborn. The charts below illustrate transaction mix, sales performance, customer ratings and relationships between numeric variables.

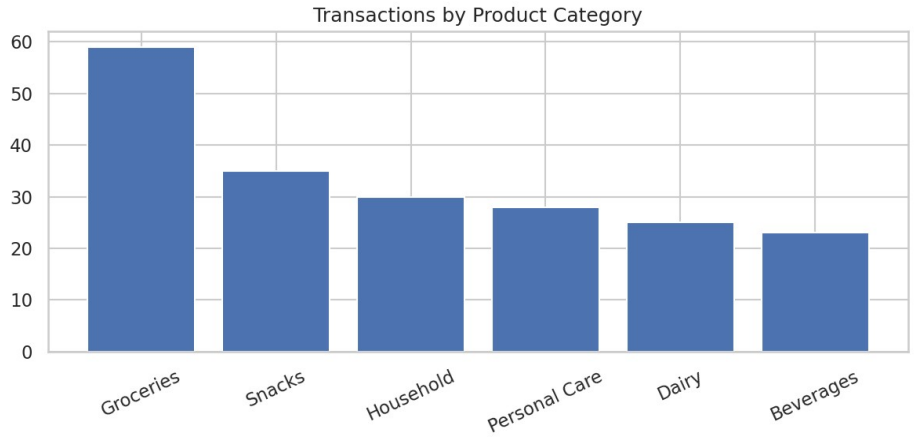


Figure 2.1: Transactions by product category (bar chart)

Result: Groceries has the most transactions (59) in the generated sample dataset.

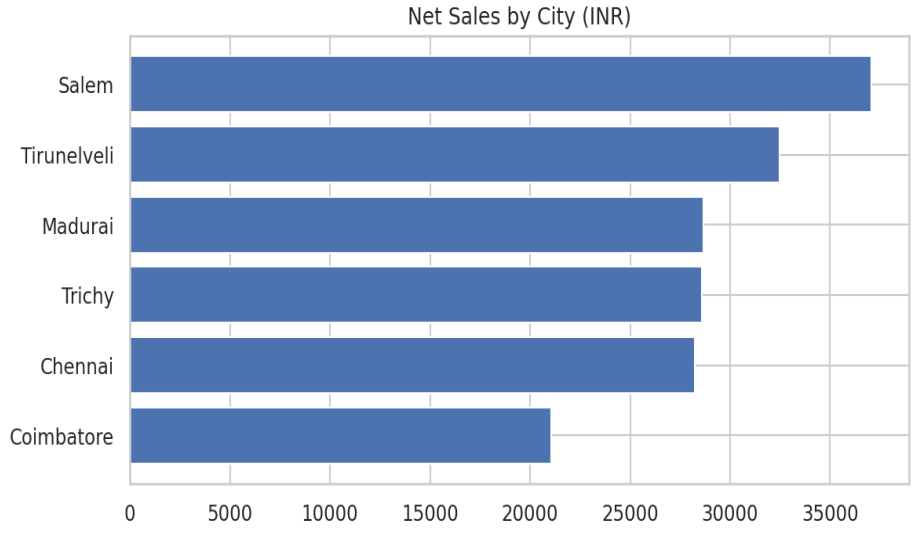


Figure 2.2: Net sales by city (horizontal bar chart)

Result: Salem records the highest net sales (₹37,071) in this sample; Coimbatore records the lowest (₹21,049).

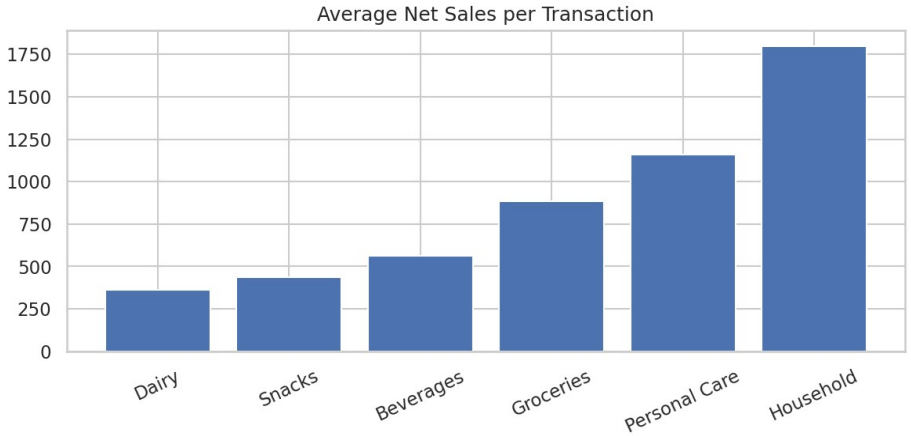


Figure 2.3: Average net sales per transaction by category

Result: Average transaction value differs by product category because units and unit prices vary.

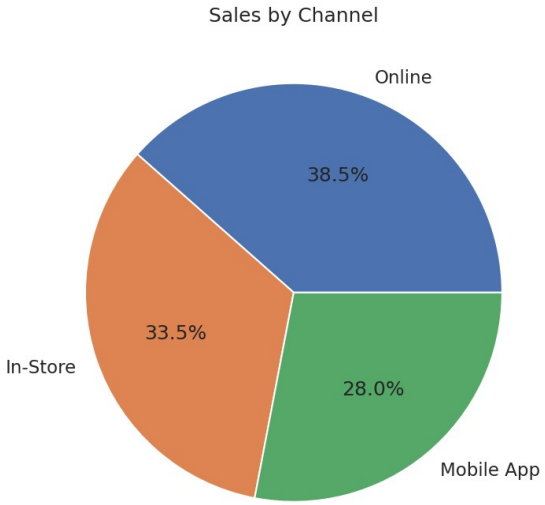


Figure 2.4: Sales by channel (pie chart)

Result: In-Store, Online and Mobile App transactions contribute to the overall sales mix.

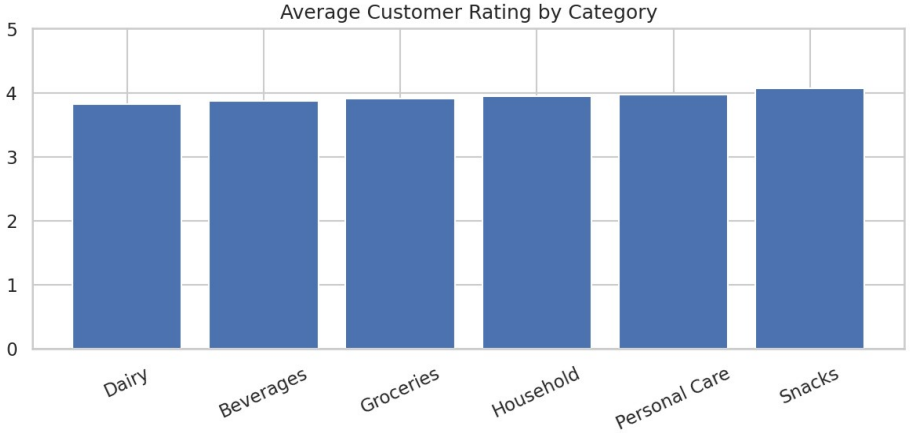


Figure 2.5: Average customer rating by category

Result: Average customer rating is approximately 3.94 out of 5 across the cleaned records.

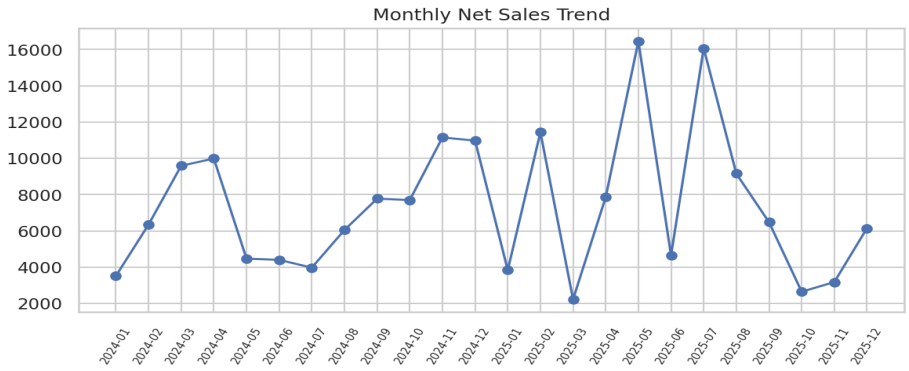


Figure 2.6: Monthly net sales trend (line chart)

Result: Monthly net sales fluctuate across the observed period; peaks and dips can be explored using dashboard filters.

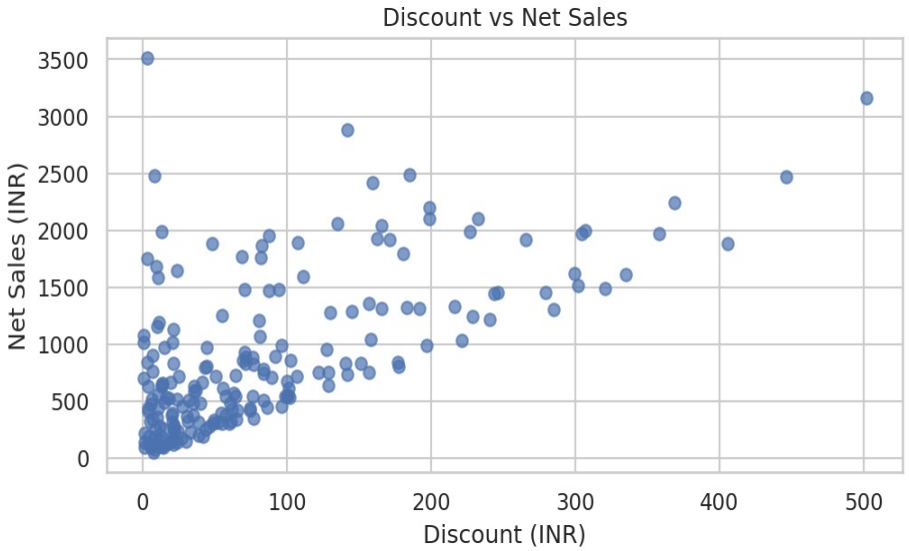


Figure 2.7: Discount vs net sales (scatter plot)

Result: The scatter plot helps inspect whether larger discounts tend to accompany larger net sales values.

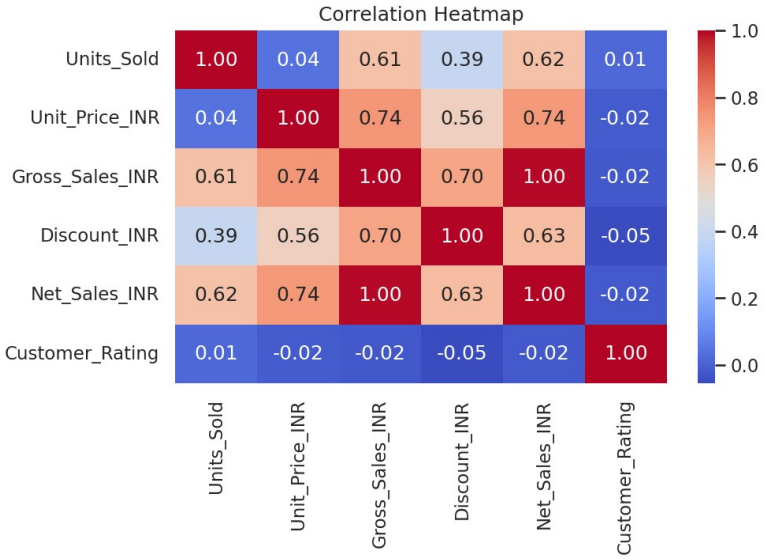


Figure 2.8: Correlation heatmap

Result: Gross sales and net sales are expected to be closely related; discount and units sold may also influence the observed relationships.

Overall, the sample illustrates differences in category demand, city-level revenue and monthly performance. These views can help retailers identify sales patterns and focus further investigation.

3. Power BI Dashboard

Power BI can be used to build a one-page interactive dashboard from Cleaned_Retail_Sales.csv.

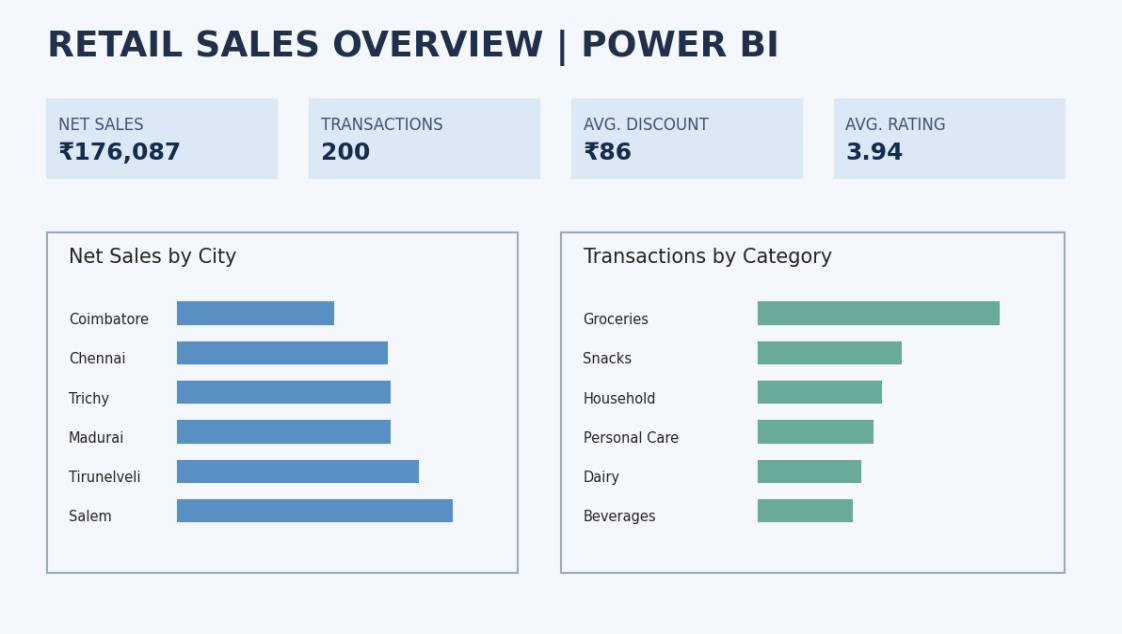


Figure 3.1: Retail Sales Dashboard in Power BI

- **Slicers: Product_Category, City, Sales_Channel and Payment_Method can filter the report; a Year slicer supports time-based comparison.**
- **KPI cards: Total Net Sales ₹176,087, Transactions 200, Average Discount ₹86 and Average Rating 3.94.**
- **Charts: category-wise transaction count, net sales by city, monthly net sales trend, and sales by channel.**
- **Insights: Groceries leads by transaction count; city totals differ, so city and channel filters can be used to investigate performance.**
- **Note: the dashboard image is a layout illustration. Recreate the visuals in Power BI using the supplied cleaned CSV to make the filters interactive.**

4. Tableau Dashboard

Tableau can be used to build a “Retail Store Performance & Business Insights Dashboard” from the same cleaned data.

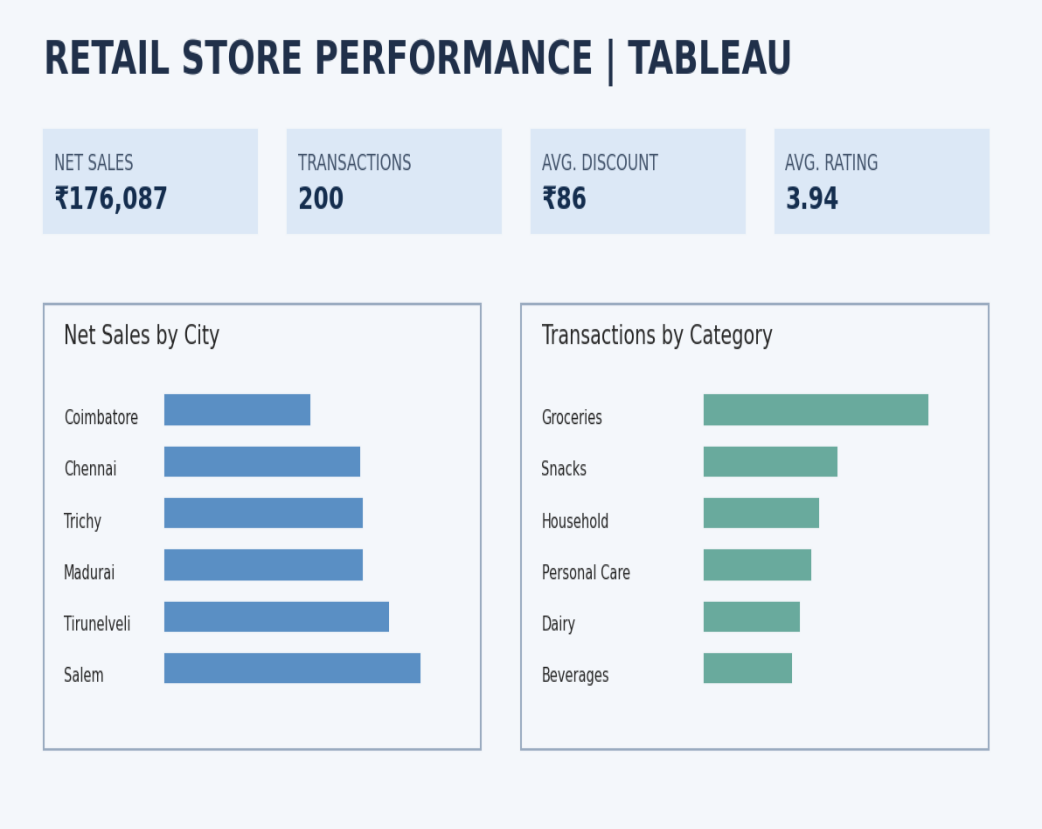


Figure 4.1: Retail Sales Dashboard in Tableau

- **Filters: City, Product Category, Sales Channel and Payment Method filters, plus a Year filter, allow focused comparisons.**
- **KPIs: Total Net Sales ₹176,087, Transactions 200, Average Discount ₹86 and Average Sales Value ₹880.**
- **Charts: monthly sales trend, sales by product category, sales channel share, net sales by city and customer ratings by category.**
- **Insights: city and product-category performance can be compared to identify high-performing locations and product groups. The supplied dashboard is a visual layout guide.**